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Jörg Barkoczi

Education

Computer Science M.Sc., *RWTH Aachen University*, Aachen. **2025–today**
Expected graduation: 2028

Mathematics B.Sc., *RWTH Aachen University*, Aachen. **2022–today**
Second degree, studied in parallel

Computer Science B.Sc., *RWTH Aachen University*, Aachen. **2022–2025**
Grade: 1.2, with distinction (1.0 = best), completed in standard time

Abitur, *Lessing-Gymnasium und -Berufskolleg*, Düsseldorf. **2019–2022**
Grade: 1.0

Publications

2026: Zijian Yang, **Jörg Barkoczi**, Ralf Schlüter, and Hermann Ney. *Sequence-Level Unsupervised Training in Speech Recognition: A Theoretical Study*. In *Proc. IEEE ICASSP*, Barcelona, Spain, May 2026. [arXiv] [IEEE]

Experience

Student Research Assistant, *Chair of Human Language Technology and Pattern Recognition (i6), RWTH Aachen University*, **2025–today**
Supervisor: Prof. Hermann Ney.
Research on unsupervised and generative training for automatic speech recognition.

- Co-authored an ICASSP 2026 paper on sequence-level unsupervised training; implemented simulations to empirically validate the theoretical results.
- Trained Conformer CTC acoustic models on LibriSpeech in PyTorch, developing generative training objectives, GMM-alignment warm-up and iterative soft-target CTC.
- Built a pipeline from self-supervised wav2vec 2.0 features (PCA, Gaussian HMM) to RASR decoding, with frame-level accuracy analysis.

Student Tutor, *RWTH Aachen University*. **2023–2025**
Weekly tutorials for groups of up to 60 students in *Discrete Structures* (Chair of Algebra and Number Theory, 2023–2024) and *Computability and Complexity* (Chair of Algorithms and Complexity, 2024–2025).

Projects

2026: **Selective Audio-Visual Speech Recognition** – Deep Learning Lab, RWTH Aachen (team of 4). Routes utterances to an audio-visual lip-reading model only when the video is predicted to help, improving on Whisper alone while invoking it on only 10% of utterances. My part: fine-tuned the audio-visual Conformer (AVEC) on in-the-wild video and built the decoding pipeline (CTC beam search, 4-gram LM, RoBERTa reranking).

Skills

Programming: Python, C++, Rust, Linux

Frameworks: PyTorch, NumPy, RETURNN, Sisyphus

Honors and Awards

2025: **Dean's List for the entire B.Sc. degree**, RWTH Aachen University – top 5%

2023–2025: **Dean's List** in every academic year – top 5% of students in my year

Languages

German (native), English (proficient), Portuguese (intermediate), Spanish (basic)